

“CipherCampus” - Secure Academic Collaboration & Anonymous Reporting Platform

A privacy-first university platform where students can:

- Share notes and academic posts securely
- Send encrypted messages
- Report issues anonymously
- Store personal academic documents safely
- Secure social network

Main Modules

1. Authentication System

- Registration + Login
- Password hashing + salting
- 2FA (OTP via email)

2. Key Management Module

- Generate RSA keys → for user data & documents
- Generate ECC keys → for messaging
- Store encrypted private keys
- Rotate keys periodically

3. Secure Academic Feed (Posts)

Users can:

- Create posts (notes, resources)
- Edit/delete posts

Security:

- Posts encrypted using RSA
- Integrity verified using HMAC

4. Encrypted Messaging System

- One-to-one chat between users

Security:

- ECC encryption
- HMAC for integrity

5. Anonymous Reporting

Users can:

- Report harassment, unfair grading, etc.

Security:

- Identity hidden using encryption
- Only admin can decrypt reports

6. Secure Document Vault

Users can:

- Upload academic files (certificates, assignments)

Security:

- Fully encrypted using RSA
- Even database admin can't read it

7. Role-Based Access Control (RBAC)

Roles:

- Admin

- View reports
- Manage users
- Student/User
 - Post, message, upload files

8. Session Management

- Secure cookies / tokens
- Prevent session hijacking

Cryptography Mapping

Feature	Algorith m
User data	RSA
Document s	RSA
Messaging	ECC
Key exchange	RSA
Integrity	HMAC
Passwords	Hash + Salt

Key Rotation Example

- Every 30 days OR login
- Re-encrypt:
 - Posts
 - Messages
 - Documents

Simple User Flow

1. User registers → keys generated
2. Logs in with 2FA
3. Creates encrypted post
4. Sends encrypted message
5. Uploads encrypted document
6. (Optional) submits anonymous report